

Quantitative Portfolio Risk and Asset Pricing Models
Center for Financial Engineering and Asset Management

1. The Black-Scholes Option Pricing PDE

Under geometric Brownian motion for underlying asset price S , the European call price C satisfies:

$$\frac{dC}{dt} + \frac{1}{2} \sigma^2 S^2 \left(\frac{d^2 C}{dS^2} \right) + r S \left(\frac{dC}{dS} \right) - r C = 0$$

where σ is implied volatility and r is the risk-free continuously compounded interest rate.

2. Modern Portfolio Theory and Sharpe Ratio

The Sharpe ratio S_p measures excess return per unit of portfolio standard deviation:

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma_p$$

Table 1: Asset class performance metrics across a 10-year investment horizon.

Asset Class	Annual Return	Annual Volatility	Sharpe Ratio
US Large Cap	12.4%	15.2%	0.68
Emerging Mkts	8.1%	22.4%	0.27
US Treasuries	3.2%	4.1%	0.29
Commodities	5.8%	18.6%	0.20

Figure 1 (Page 1): The Efficient Frontier curve with capital allocation line tangent at the optimal tangency portfolio.